

Agentic AI for Sustainable Hospitality

This PDF is a page-by-page export of my Excel workbook, the measurement model behind the case study: waste, forecast accuracy, guest experience, AgentOps, the readiness gate and economics.

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MIT • Implementing Agentic AI program

HOW TO READ THIS PDF

Each page is one tab of the workbook (Hotel_Food_Waste_Model.xlsx), exported as it appears in Excel. Formulas are shown as their calculated values. The wide strategy tab is split into one page per section. The tab name is at the bottom right of every page.

13 TABS

MEASURE

Weekly Waste • Waste by Source • Forecast MAPE • Waste Cost • Guest Experience • Operations

GOVERN

AgentOps • Readiness Gate (8 thresholds, all must pass) • Control Room • Overrides

DECIDE

Economics • Sensitivity

EXECUTE

Strategy execution (1 tab, 7 sections): strategy map • scorecard • OKR view • performance view • execution matrix • operating model

Headline results (modeled: 4 baseline weeks vs 8 agent weeks)

Weekly food waste 489 → 304 kg (−37.8%, baseline avg vs final 4 weeks)

Forecast MAPE 17.8% → 7.2% (baseline avg vs week A8)

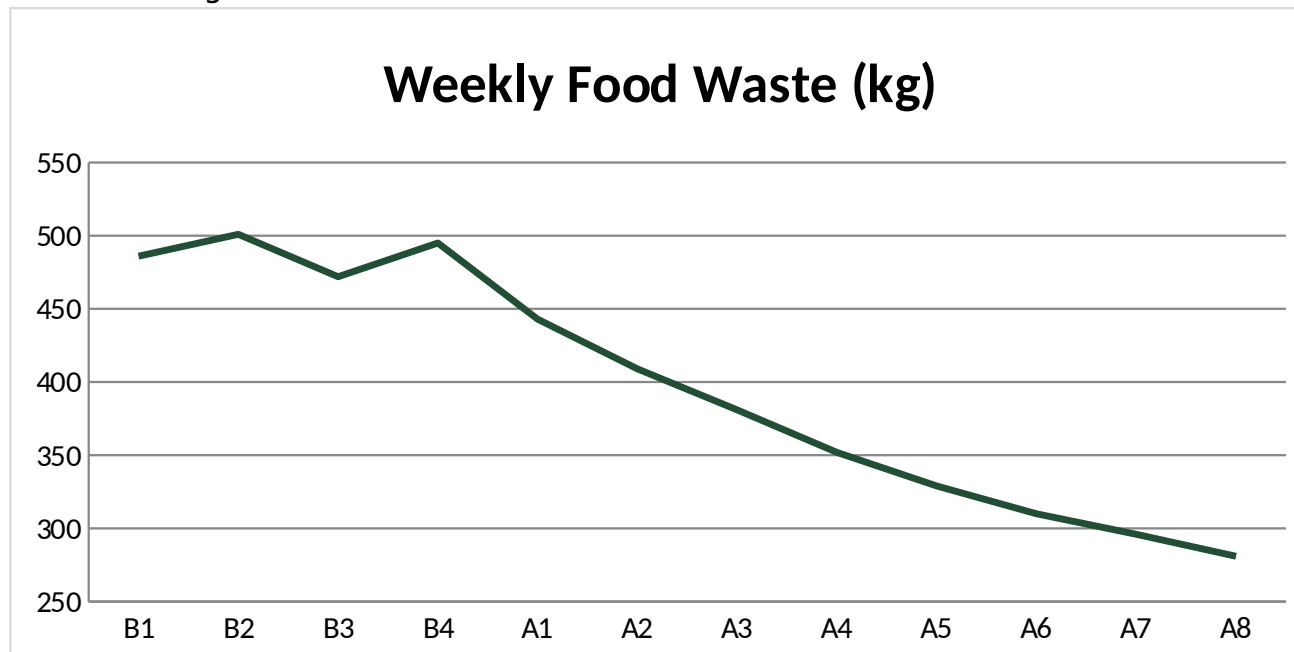
Guest F&B score 4.61 → 4.68 (B1 vs A8)

Readiness gate HOLD (A1–A6) → PASS (A7–A8)

All figures are modeled for the case study. No live hotel systems or real hotel data.

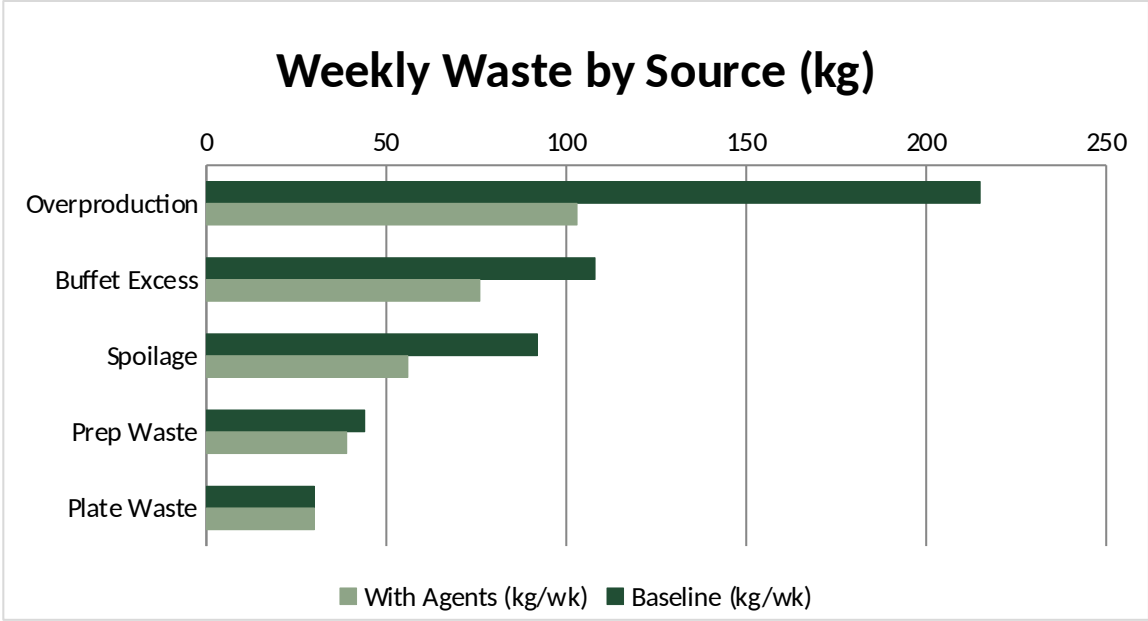
Week	Phase	Food Waste (kg)
B1	Baseline	486
B2	Baseline	501
B3	Baseline	472
B4	Baseline	495
A1	Agent	443
A2	Agent	409
A3	Agent	381
A4	Agent	352
A5	Agent	329
A6	Agent	310
A7	Agent	296
A8	Agent	281

Baseline Avg (B1-B4)	489
Final 4-Wk Avg (A5-A8)	304
Change	-37.8%



Waste Source	Baseline (kg/wk)	With Agents (kg/wk)	Change
Overproduction	215	103	-52%
Buffet Excess	108	76	-30%
Spoilage	92	56	-39%
Prep Waste	44	39	-11%
Plate Waste	30	30	0%
Total	489	304	-38%

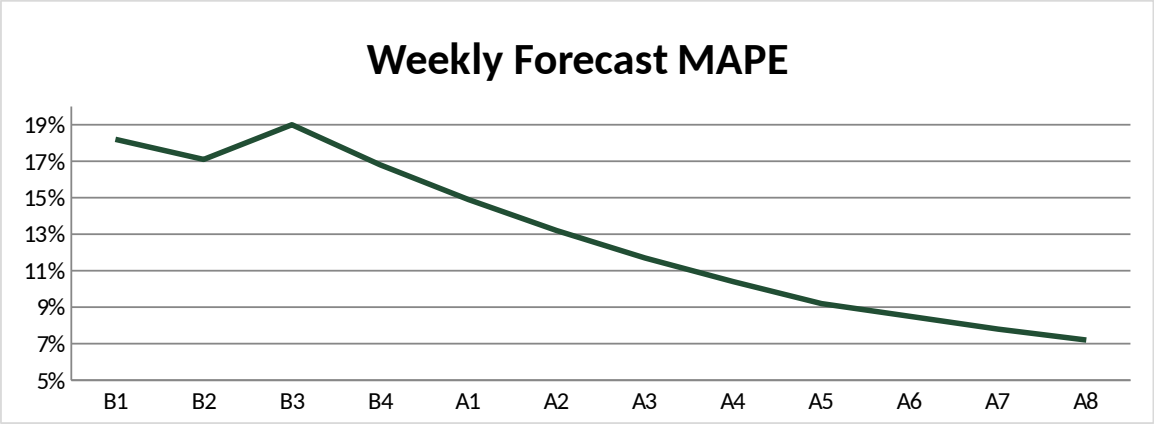
Baseline = B1-B4 average; With Agents = A5-A8 average



Week	Phase	Forecast MAPE
B1	Baseline	18.2%
B2	Baseline	17.1%
B3	Baseline	19.0%
B4	Baseline	16.8%
A1	Agent	14.9%
A2	Agent	13.2%
A3	Agent	11.7%
A4	Agent	10.4%
A5	Agent	9.2%
A6	Agent	8.5%
A7	Agent	7.8%
A8	Agent	7.2%

Baseline Avg (B1-B4)	17.8%
Final Week (A8)	7.2%
Change	-59%

MAPE = mean(|Actual - Forecast| / Actual) x 100, across the week's services.



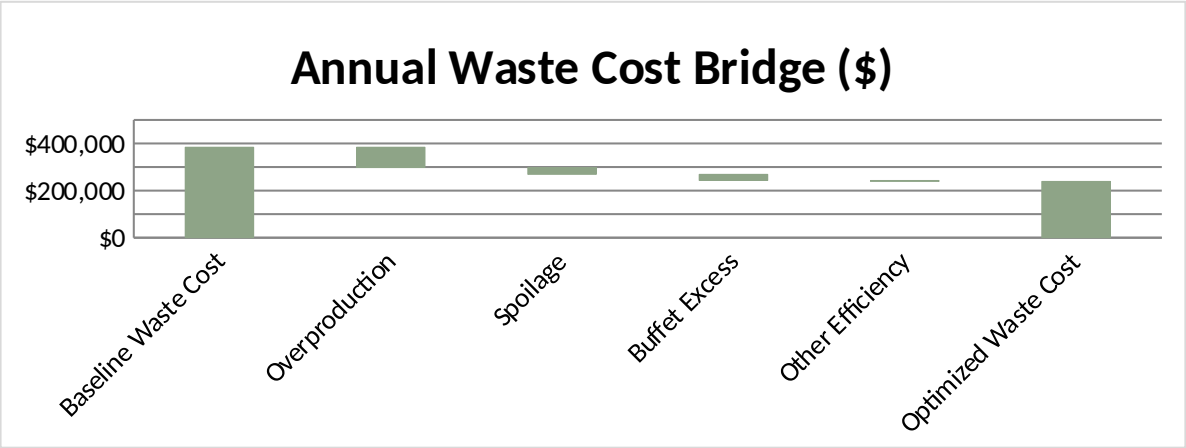
Step	Amount (\$)
Baseline Waste Cost	\$384,000
Overproduction	(\$86,000)
Spoilage	(\$29,000)
Buffet Excess	(\$26,000)
Other Efficiency	(\$4,000)
Optimized Waste Cost	\$239,000

Annual Food Cost Protected
Reduction in Waste Cost

\$145,000
37.8%

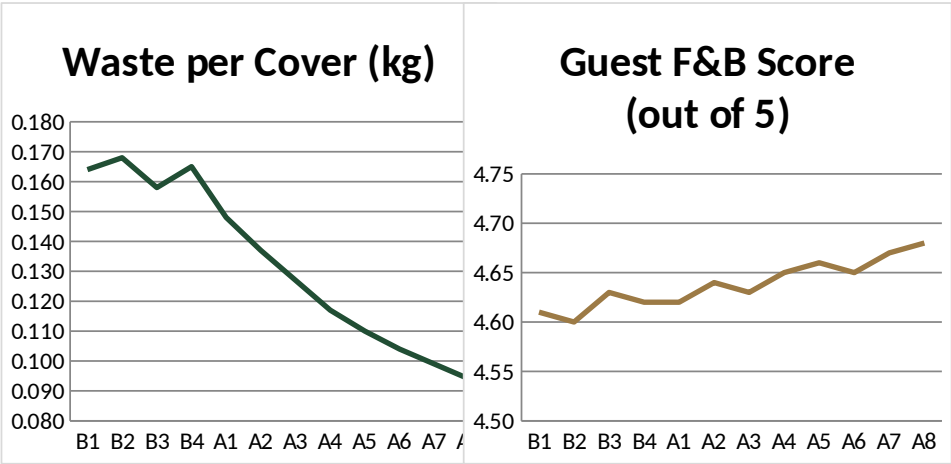
Avoided waste cost, not incremental revenue.

Inputs	
Annual Food Purchasing	\$3,200,000
Baseline Annual Waste Cost	\$384,000
Waste Cost % of Purchasing	12.0%

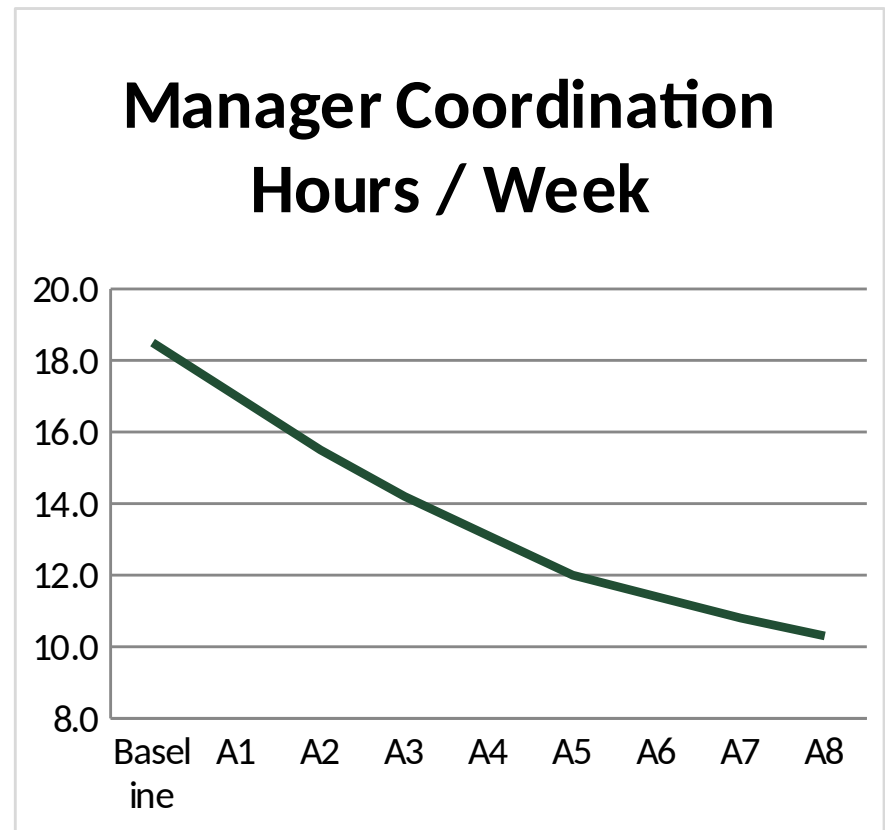


Week	Phase	Waste / Cover (kg)	Guest F&B Score
B1	Baseline	0.164	4.61
B2	Baseline	0.168	4.60
B3	Baseline	0.158	4.63
B4	Baseline	0.165	4.62
A1	Agent	0.148	4.62
A2	Agent	0.137	4.64
A3	Agent	0.127	4.63
A4	Agent	0.117	4.65
A5	Agent	0.110	4.66
A6	Agent	0.104	4.65
A7	Agent	0.099	4.67
A8	Agent	0.094	4.68

Waste / Cover: Baseline Avg	0.164
Waste / Cover: Week A8	0.094
Waste / Cover Change	-43%
Guest Score: B1	4.61
Guest Score: A8	4.68



Week	Coordination Hours / Week
Baseline	18.5
A1	17.0
A2	15.5
A3	14.2
A4	13.1
A5	12.0
A6	11.4
A7	10.8
A8	10.3
Baseline	18.5
Week A8	10.3
Change	-44%



Week	Recommendations	Accepted	Human Overrides	Required Escalations	Correct Escalations	All Escalations	Tool Calls	Successful Tool Calls	Acceptance Rate	Override Rate	Escalation Recall	Escalation Precision	Tool Reliability	Autonomy Score	
A1		100	61	29	27	20	27	520	503	61.0%	29.0%	74.1%	74.1%	96.7%	73.2
A2		108	72	27	27	21	27	545	531	66.7%	25.0%	77.8%	77.8%	97.4%	77.1
A3		115	83	24	28	23	28	570	559	72.2%	20.9%	82.1%	82.1%	98.1%	81.2
A4		121	92	22	27	23	27	595	586	76.0%	18.2%	85.2%	85.2%	98.5%	84.1
A5		128	102	19	26	23	27	620	612	79.7%	14.8%	88.5%	85.2%	98.7%	86.4
A6		132	110	17	25	23	26	640	633	83.3%	12.9%	92.0%	88.5%	98.9%	89.2
A7		139	120	15	27	25	27	665	659	86.3%	10.8%	92.6%	92.6%	99.1%	91.4
A8		145	128	13	22	21	23	690	686	88.3%	9.0%	95.5%	91.3%	99.4%	92.6

Source: Hotel_Food_Waste_AgentOps_Metrics.xlsx (Amelia Wolfire) — values and formulas reproduced unchanged.

Readiness Criterion	Test	Threshold
Recommendation Acceptance	≥	80% <i>Illustrative thresholds — edit the blue cells.</i>
Escalation Recall	≥	90%
Escalation Precision	≥	85%
Tool Reliability	≥	99%
Human Override Rate	≤	15%
Forecast MAPE	≤	10%
Guest F&B Score	≥	4.60
Critical Policy Violations	=	0

Week	Acceptance	Recall	Precision	Tool Reliability	Override Rate	MAPE	Guest Score	Policy Violations	Failed Checks	Gate	First Blocker	Autonomy Score
A1	61.0%	74.1%	74.1%	96.7%	29.0%	14.9%	4.62	0	6	HOLD	Acceptance	73.2
A2	66.7%	77.8%	77.8%	97.4%	25.0%	13.2%	4.64	0	6	HOLD	Acceptance	77.1
A3	72.2%	82.1%	82.1%	98.1%	20.9%	11.7%	4.63	0	6	HOLD	Acceptance	81.2
A4	76.0%	85.2%	85.2%	98.5%	18.2%	10.4%	4.65	0	5	HOLD	Acceptance	84.1
A5	79.7%	88.5%	85.2%	98.7%	14.8%	9.2%	4.66	0	3	HOLD	Acceptance	86.4
A6	83.3%	92.0%	88.5%	98.9%	12.9%	8.5%	4.65	0	1	HOLD	Tool reliability	89.2
A7	86.3%	92.6%	92.6%	99.1%	10.8%	7.8%	4.67	0	0	PASS	—	91.4
A8	88.3%	95.5%	91.3%	99.4%	9.0%	7.2%	4.68	0	0	PASS	—	92.6

Policy violations: modeled at 0 for every agent week (illustrative).

Autonomy Score construction

Recommendation Acceptance	30%
Escalation Recall	25%
Escalation Precision	20%
Tool Reliability	15%
Inverse Override Rate (1 – override)	10%
Weight total	100%

Case-study composite index; illustrative weighting. A high score informs, but never by itself grants, more authority.

Category	KPI (week A8)	Value	Test	Target	Status	Source
QUALITY	Recommendation Acceptance	88.3%	>=	80.0%	On target	AgentOps
QUALITY	Forecast MAPE	7.2%	<=	10.0%	On target	Forecast MAPE
QUALITY	Recommendation Outcome Quality	91.0%	>=	85.0%	On target	<i>Modeled</i>
SAFETY	Escalation Recall	95.5%	>=	90.0%	On target	AgentOps
SAFETY	Escalation Precision	91.3%	>=	85.0%	On target	AgentOps
SAFETY	Policy Violations	0	=	0	On target	<i>Modeled</i>
SAFETY	Correct Abstention Rate	92.0%	>=	90.0%	On target	<i>Modeled</i>
RELIABILITY	Tool / API Reliability	99.4%	>=	99.0%	On target	AgentOps
RELIABILITY	Context Completeness	97.8%	>=	95.0%	On target	<i>Modeled</i>
RELIABILITY	Human Override Rate	9.0%	<=	15.0%	On target	AgentOps
RELIABILITY	Execution Failure Rate	0.6%	<=	1.0%	On target	<i>Modeled</i>
BUSINESS	Food Waste Reduction	37.8%			Outcome	Weekly Waste
BUSINESS	Waste Cost Avoided (annual)	\$145,000			Outcome	Waste Cost
BUSINESS	Manager Hours Saved / Week	8.2			Outcome	Operations
BUSINESS	Guest F&B Score	4.68	>=	4.60	On target	Guest Experience

Blue values marked 'Modeled' are illustrative assumptions for the case study, not pilot measurements.

Override Reason	Overrides (A1-A8)	Share	System Improvement
Missing context	38	23%	Add banquet and group-block feed to context layer
Special event	31	19%	Integrate events calendar; lower autonomy on event days
Forecast disagreement	29	17%	Retrain demand model on chef-adjusted actuals
Guest-experience concern	24	14%	Add minimum-display guardrail for signature items
Inventory constraint	19	11%	Refresh inventory feed hourly; block on stale data
Policy requirement	14	8%	Encode the policy as a hard constraint
Agent error	11	7%	Log as defect; add a regression case to the evaluation suite
Total	166		
Check vs AgentOps overrides	166	Reconciles	
<i>Reason split is a modeled diagnostic tagging of the 166 pilot overrides (illustrative).</i>			

Drift Scenario (illustrative, post- t)	Override Rate	Trailing 3-wk A	Drift Signal
P1	9%		
P2	10%		
P3	11%		
P4	23%	10%	DRIFT DETECTED
<i>Rule: flag drift when the week's override rate exceeds 1.5× its trailing 3-week average.</i>			

Line Item	Annual (\$)	Basis
VALUE		
Food cost avoided (incl. spoilage + inventory utilization)	\$145,000	Waste Cost sheet
Coordination capacity redeployed	\$19,188	Hours saved × 52 × loaded rate
Gross annual benefit	\$164,188	
AGENTIC AI OPERATING COST (annual)		
Model / API usage	\$6,000	Modeled assumption (illustrative) — edit to test.
Monitoring + observability	\$6,000	Modeled assumption (illustrative) — edit to test.
Human review of approvals	\$4,680	Review hours × 52 × loaded rate
Maintenance + evaluation	\$12,000	Modeled assumption (illustrative) — edit to test.
Annual AI operating cost	\$28,680	
ONE-TIME COST		
Integration (PMS, POS, inventory, procurement)	\$40,000	Modeled assumption (illustrative) — edit to test.
Training + change management	\$10,000	Modeled assumption (illustrative) — edit to test.
Total one-time cost	\$50,000	
INPUTS		
Manager loaded hourly cost (\$)	\$45	Modeled assumption (illustrative) — edit to test.
Manager hours saved / week	8.2	
Human review hours / week	2.0	Modeled assumption (illustrative) — edit to test.
RESULTS		
Net annual benefit (run-rate)	\$135,508	
Year-1 net benefit	\$85,508	
Year-1 ROI	109%	
Payback period (months)	4.4	
Value / AI operating cost	5.7x	

Coordination capacity is redeployed manager time, not a cash saving.

Scenario Driver	Conservative	Base	Upside
Waste reduction	15%	25%	38%
Adoption rate	80%	90%	100%
AI operating-cost multiplier	125%	100%	100%
Food purchasing volume index	100%	100%	100%

RESULTS

Food cost avoided	\$46,080	\$86,400	\$145,000
Coordination capacity	\$15,350	\$17,269	\$19,188
Gross benefit	\$61,430	\$103,669	\$164,188
AI operating cost	\$35,850	\$28,680	\$28,680
Net annual benefit	\$25,580	\$74,989	\$135,508
Payback (months)	23.5	8.0	4.4

One-way sensitivity (Base net benefit)	Low Case	High Case	Swing
Waste reduction (15% → Upside)	\$40,429	\$119,089	\$78,660
Adoption rate (70% → 100%)	\$51,952	\$86,508	\$34,556
Food purchasing volume (−20% → +20%)	\$57,709	\$92,269	\$34,560
AI operating cost (×1.5 → ×0.5)	\$60,649	\$89,329	\$28,680

Forecast improvement is expressed through the waste-reduction driver. All scenario inputs are modeled assumptions.

10 · STRATEGY EXECUTION & PERFORMANCE MANAGEMENT

What does leadership actually do with these metrics?

INSIGHT → DECISION → OWNER → ACTION → MEASUREMENT → STRATEGIC OUTCOME				
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Governance review weekA6Select A1-A8 from the drop-down. Drives Sections A, B and D and the AI-governance row in F. A6 is the week the readiness gate held; switch to A8 for the end-of-pilot review.

All values are calculated in this workbook from modeled case-study data (12-week pilot scenario), not realized hotel performance.

A · STRATEGY MAP

Four strategic objectives in causal order, with AI governance as the enabler beneath them. Each objective's status rolls up automatically from the Section B scorecard: its weakest KPI sets the status.

Level	Strategic Objective	KPIs (Section B)	Status	Leadership Question
OBJECTIVE 1	Improve Decision Intelligence	Forecast MAPE · Recommendation Acceptance · Context Completeness	ON TRACK	Do we know what demand to expect?
↓				
OBJECTIVE 2	Improve Operational Efficiency	Overproduction · Waste per Cover · Food Waste · Manager Hours Saved	AT RISK	Are we producing only what guests will use?
↓				
OBJECTIVE 3	Protect Guest Value	Guest F&B Score · Guest Experience Guardrail · Service-Related Exceptions	ON TRACK	Is the guest experience protected?
↓				
OBJECTIVE 4	Create Sustainable Business Value	Food Cost Avoided · Net Annual Benefit · ROI · Payback Period	ON TRACK	Is it worth it, net of the AI's own cost?
ENABLER	Earn Greater AI Autonomy	Reliability · Escalation Recall · Override Rate · Readiness Gate	OFF TRACK	Has the agent earned more decision authority?

CAUSAL LOGIC: BETTER DECISION INTELLIGENCE → BETTER OPERATING DECISIONS → LOWER RESOURCE WASTE → GUEST EXPERIENCE PROTECTED → MEASURABLE BUSINESS VALUE

B · EXECUTIVE SCORECARD											
Period: governance review week A6 (cell B4). Weekly metrics read that week; pilot-level and financial metrics are fixed pilot results. Status and Required Action calculate automatically.											
Strategic Objective	KPI	Baseline	Current	Target	Variance	Trend	Status	Owner	Review Cadence	Strategic Initiative	Required Action
Improve Decision Intelligence	Forecast MAPE	17.8%	8.5%	≤ 10.0%	-1.5%▲	Improving	ON TRACK	AI / Product	Weekly	Improve demand forecasting	Continue — review weekly
Improve Decision Intelligence	Recommendation Acceptance	61.0%	83.3%	≥ 80.0%	+3.3%▲	Improving	ON TRACK	Operations	Weekly	Introduce governed production recommendations	Continue — review weekly
Improve Decision Intelligence	Context Completeness (A8, modeled)		97.8%	≥ 95.0%	+2.8%—		ON TRACK	IT / Data	Daily	Connect inventory context	Continue — review daily
Improve Operational Efficiency	Weekly Food Waste (kg)	489	310	≤ 342	-32▲	Improving	ON TRACK	F&B Leadership	Weekly	Monitor waste root causes	Continue — review weekly
Improve Operational Efficiency	Overproduction (kg / week, final 4-wk avg)	215	103	≤ 120	-17▲	Improving	ON TRACK	Operations	Weekly	Introduce governed production recommendations	Continue — review weekly
Improve Operational Efficiency	Waste per Cover (kg)	0.164	0.104	≤ 0.115	-0.011▲	Improving	ON TRACK	F&B Leadership	Weekly	Monitor waste root causes	Continue — review weekly
Improve Operational Efficiency	Manager Hours Saved / Week	0.0	7.1	≥ 8.0	-0.9▲	Improving	AT RISK	Operations	Monthly	Introduce governed production recommendations	Investigate — remove manual re-work from the approval flow
Protect Guest Value	Guest F&B Score	4.61	4.65	≥ 4.60	+0.05▲	Improving	ON TRACK	F&B Leadership	Weekly	Operate readiness gate	Continue — review weekly
Protect Guest Value	Guest Experience Guardrail (% of agent weeks ≥ score floor)		100%	≥ 100%	0%—		ON TRACK	Risk / Governance	Weekly	Operate readiness gate	Continue — review weekly
Protect Guest Value	Service-Related Exceptions / Week (modeled)	5	2	≤ 3	-1▲	Improving	ON TRACK	Operations	Weekly	Connect inventory context	Continue — review weekly
Create Sustainable Business Value	Food Cost Avoided (annual)		\$145,000	≥ \$100,000	+\$45,000—		ON TRACK	F&B Leadership	Quarterly	Agent-enabled demand and production decision system	Continue — review quarterly
Create Sustainable Business Value	Net Annual Benefit (after AI cost)		\$135,508	≥ \$74,989	+\$60,519—		ON TRACK	F&B Leadership	Quarterly	Agent-enabled demand and production decision system	Continue — review quarterly
Create Sustainable Business Value	Year-1 ROI		109%	≥ 50%	+59%—		ON TRACK	F&B Leadership	Quarterly	Agent-enabled demand and production decision system	Continue — review quarterly
Create Sustainable Business Value	Payback (months)		4.4	≤ 12.0	-7.6—		ON TRACK	F&B Leadership	Quarterly	Agent-enabled demand and production decision system	Continue — review quarterly
Earn Greater AI Autonomy	Tool / System Reliability	96.7%	98.9%	≥ 99.0%	-0.1%▲	Improving	OFF TRACK	IT / Data	Weekly	Operate readiness gate	Hold Autonomy — fix failing integrations before any authority review
Earn Greater AI Autonomy	Escalation Recall	74.1%	92.0%	≥ 90.0%	+2.0%▲	Improving	ON TRACK	Risk / Governance	Weekly	Review human overrides	Continue — review weekly
Earn Greater AI Autonomy	Human Override Rate	29.0%	12.9%	≤ 15.0%	-2.1%▲	Improving	ON TRACK	AI / Product	Weekly	Review human overrides	Continue — review weekly
Earn Greater AI Autonomy	Production Readiness Gate (all 8 criteria)	HOLD	HOLD	PASS	—	► Flat	OFF TRACK	Risk / Governance	Weekly	Operate readiness gate	Hold Autonomy — resolve first blocker: Tool reliability
Status: ON TRACK = meets target · AT RISK = misses target but within tolerance (col O) · OFF TRACK = outside tolerance. Readiness-gate criteria use zero tolerance. Trend compares Current with Baseline (baseline for agent-only metrics = week A1). Blue = editable illustrative targets and tolerances; green = linked from other sheets; black = calculated.											

CALCULATION HELPERS · blue = editable				
Direction	Tolerance	Data Source	Action if not on track	Display format
Lower	0.0%	Forecast MAPE	retrain the demand model on chef-adjusted actuals	0.0%
Higher	0.0%	AgentOps	review rejected recommendations with chefs	0.0%
Higher	0.0%	Control Room (modeled)	restore missing data feeds before execution	0.0%
Lower	25	Weekly Waste	run a root-cause review on the top waste category	#,##0
Lower	10	Waste by Source	tighten prep ranges on chronic items	#,##0
Lower	0.010	Guest Experience	review portioning and buffet refill rules	0.000
Higher	1.5	Operations	remove manual re-work from the approval flow	0.0
Higher	0.00	Guest Experience	pause reductions on affected menu items	0.00
Higher	0%	Guest Experience	reduce autonomy and investigate guest impact	0%
Lower	1	Modeled input (illustrative)	review stockouts against production cuts	0
Higher	\$20,000	Waste Cost	re-baseline purchasing assumptions	\$#,##0
Higher	\$15,000	Economics · Sensitivity	revisit AI operating-cost assumptions	\$#,##0
Higher	25%	Economics	phase integration spend	0%
Lower	6.0	Economics	phase integration spend	0.0
Higher	0.0%	AgentOps · Readiness Gate	fix failing integrations before any authority review	0.0%
Higher	0.0%	AgentOps	audit missed escalations	0.0%
Lower	0.0%	AgentOps · Overrides	diagnose override reasons and retrain	0.0%
Gate	—	Readiness Gate	resolve the first blocker	@

C · PERDOO-STYLE STRATEGY / OKR VIEW

OBJECTIVE	Reduce avoidable food waste without compromising guest experience.					100%	ON TRACK	F&B Leadership		100%	
#	Key Result	KR Type	Start	Target	Current	Progress	Status	Owner		Progress Indicator	Initiatives
KR 1	Reduce modeled weekly food waste by ≥ 30%	Decrease by	0%	≤ -30%	-37.8%	100%	ON TRACK	F&B Leadership		100%	Monitor waste root causes · Introduce governed production recommendations
KR 2	Maintain Guest F&B Score ≥ 4.60	Maintain ≥	4.61	≥ 4.60	4.68	100%	ON TRACK	F&B Leadership		100%	Operate readiness gate
KR 3	Maintain Forecast MAPE ≤ 10%	Maintain ≤	17.8%	≤ 10%	7.2%	100%	ON TRACK	AI / Product		100%	Improve demand forecasting
KR 4	Maintain total system reliability ≥ 99%	Maintain ≥	96.7%	≥ 99%	99.4%	100%	ON TRACK	IT / Data		100%	Operate readiness gate · Connect inventory context

#	Initiative	Linked Key Results					Status	Owner		Evidence (end of pilot)	What It Changes
INIT 1	Improve demand forecasting	KR 3					ON TRACK	AI / Product		MAPE 17.8% → 7.2%	Demand Agent adds reservations, events and weather
INIT 2	Connect inventory context	KR 4					ON TRACK	IT / Data		Context completeness 97.8% (modeled)	Inventory Agent reads stock, shelf life and procurement
INIT 3	Introduce governed production recommendations	KR 1					ON TRACK	Operations		Acceptance 61% → 88%	Orchestrator plans; approvals routed by risk
INIT 4	Monitor waste root causes	KR 1					ON TRACK	F&B Leadership		Overproduction -52%	Waste Agent tags weight, category and reason
INIT 5	Operate readiness gate	KR 2 · KR 4					ON TRACK	Risk / Governance		First gate PASS: week A7	8 thresholds; autonomy expands only on evidence
INIT 6	Review human overrides	KR 1 – KR 4					ON TRACK	AI / Product		Override rate 29% → 9%; 166 overrides diagnosed	Override → reason → diagnosis → system improvement

D • CLEARPOINT STRATEGY-STYLE PERFORMANCE VIEW									
Strategic Objective	Measure	Target	Actual	Variance	Status	Trend	Owner	Initiative	Management Commentary
Improve Decision Intelligence	Objective roll-up · all Section B measures				ON TRACK		AI / Product		3 of 3 measures on track in week A6.
	Forecast MAPE	≤ 10.0%	8.5%	-1.5%	ON TRACK	▲ Improving	AI / Product	Improve demand forecasting	Forecast MAPE: 8.5% vs target ≤ 10.0% — on track. Continue.
	Recommendation Acceptance	≥ 80.0%	83.3%	+3.3%	ON TRACK	▲ Improving	Operations	Introduce governed production recommendations	Recommendation Acceptance: 83.3% vs target ≥ 80.0% — on track. Continue.
Improve Operational Efficiency	Objective roll-up · all Section B measures				AT RISK		Operations		3 of 4 measures on track in week A6.
	Weekly Food Waste (kg)	≤ 342	310	-32	ON TRACK	▲ Improving	F&B Leadership	Monitor waste root causes	Weekly Food Waste (kg): 310 vs target ≤ 342 — on track. Continue.
	Manager Hours Saved / Week	≥ 8.0	7.1	-0.9	AT RISK	▲ Improving	Operations	Introduce governed production recommendations	Manager Hours Saved / Week: 7.1 vs target ≥ 8.0 — at risk. Investigate — remove manual re-work from the approval flow.
Protect Guest Value	Objective roll-up · all Section B measures				ON TRACK		F&B Leadership		3 of 3 measures on track in week A6.
	Guest F&B Score	≥ 4.60	4.65	+0.05	ON TRACK	▲ Improving	F&B Leadership	Operate readiness gate	Guest F&B Score: 4.65 vs target ≥ 4.60 — on track. Continue.
	Guest Experience Guardrail (% of agent weeks ≥ score floor)	≥ 100%	100%	0%	ON TRACK	—	Risk / Governance	Operate readiness gate	Guest Experience Guardrail (% of agent weeks ≥ score floor): 100% vs target ≥ 100% — on track. Continue.
Create Sustainable Business Value	Objective roll-up · all Section B measures				ON TRACK		F&B Leadership		4 of 4 measures on track in week A6.
	Food Cost Avoided (annual)	≥ \$100,000	\$145,000	+\$45,000	ON TRACK	—	F&B Leadership	Agent-enabled demand and production decision system	Food Cost Avoided (annual): \$145,000 vs target ≥ \$100,000 — on track. Continue.
	Net Annual Benefit (after AI cost)	≥ \$74,989	\$135,508	+\$60,519	ON TRACK	—	F&B Leadership	Agent-enabled demand and production decision system	Net Annual Benefit (after AI cost): \$135,508 vs target ≥ \$74,989 — on track. Continue.
Earn Greater AI Autonomy	Objective roll-up · all Section B measures				OFF TRACK		Risk / Governance		2 of 4 measures on track in week A6.
	Tool / System Reliability	≥ 99.0%	98.9%	-0.1%	OFF TRACK	▲ Improving	IT / Data	Operate readiness gate	Week A6 reliability missed the production-readiness threshold by 0.1 percentage points. Increased autonomy remains on HOLD pending evidence of recovery.
	Production Readiness Gate (all 8 criteria)	PASS	HOLD	—	OFF TRACK	► Flat	Risk / Governance	Operate readiness gate	Week A6: readiness gate on HOLD. First blocker: Tool reliability. Autonomy stays at its current level.

E · STRATEGY-TO-EXECUTION MATRIX									
Connects the whole case study: priority → objective → KPI → data source → target → actual → status → owner → decision → business impact. Pilot results, except AI Governance, which reads the review week in cell B4.									
Strategic Priority	Objective	KPI	Data Source	Target	Actual	Status	Owner	Decision	Business Impact
SUSTAINABILITY	Reduce preventable waste	Food waste (4-wk avg vs baseline)	Weekly Waste	≤ -30%	-37.8%	ON TRACK	F&B / Operations	Continue modeled intervention	≈ \$145,000 food cost avoided per year; lower purchasing volume
OPERATIONAL EXCELLENCE	Improve production decisions	Forecast MAPE (week A8)	Forecast MAPE	≤ 10%	7.2%	ON TRACK	AI / Product	Continue	Overproduction -52% as forecasts improve
GUEST VALUE	Protect experience	Guest F&B Score (week A8)	Guest Experience	≥ 4.60	4.68	ON TRACK	F&B Leadership	Guardrail satisfied	Efficiency gains do not come at the guest's expense
AI GOVERNANCE	Earn autonomy	Tool / system reliability (review week)	AgentOps · week A6	≥ 99%	98.9%	OFF TRACK	Risk / Governance	HOLD autonomy	Readiness gate HOLD in week A6. First blocker: Tool reliability
FINANCIAL PERFORMANCE	Create measurable value	Net annual benefit (pilot-result scenario)	Economics · Sensitivity	≥ \$74,989	\$135,508	ON TRACK	F&B Leadership	Evaluate continued investment	Payback 4.4 months · year-1 ROI 109% · target = base-case net benefit

